

LABORATORY SESSION 3

Cell and Tissue Injury and Repair (3)

Necrosis on the bench: coagulative, caseous, liquefactive, and the two gangrenes, then the tissue that repairs it all.

GROSS 7 specimens: spleen, lung, liver, appendix, small intestine, foot, tubular bone

SLIDES Normal spleen and skin, plus infarction, caseous necrosis, and granulation tissue

GOAL Recognize each pattern by eye, and describe it the way the bench expects

BUILT FROM

Pathological Experiment: Cell and Tissue Injury and Repair (3)

MBBS Pathology Laboratory curriculum, ZJU-ISM, Yiwu campus · specimen list and slide deck

Robbins Basic Pathology, 11th ed. (chapter 2, Cell Injury; chapter 3, Repair)

Enriches every specimen with the mechanism behind what the bench shows.

How to use this guide

Every specimen card follows the same order the bench expects: organ, gross or microscopic description, then diagnosis. Read the description before the diagnosis and try to name the lesion yourself; the diagnosis line is there to check your answer, not to replace the exercise.

Reading the tags

GROSS marks a naked-eye specimen observation; **MICRO** marks a microscopic one. A **LAB TECHNIQUE** box explains how to describe what you see; an **EXAM TRAP** box flags a distinction spot exams like to test.

Method

Specimen list, gross photographs, and slide micrographs extracted directly from the lab's own deck (a screenshot-based specimen viewer; interface chrome cropped out, credited image by image in `src/img/CREDITS.md`). Content checked twice against Robbins Basic Pathology 11e for the correct name, mechanism, and Latin/English terminology of every pattern.

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PART ONE

01

How to Work in the Lab

Every specimen on the bench is a lesion waiting to be named correctly. This part sets the two skills the rest of the session tests: what you are here to learn, and the fixed order in which you describe anything you see.

Purpose of this session

LEARNING OBJECTIVES, AS GIVEN

Understand the concepts of necrosis and gangrene, and grasp the morphological character of each type. Master the concept and structure of granulation tissue and fracture repair.

In practice that means two skills, tested on seven gross specimens and five slides: telling coagulative, caseous, and liquefactive necrosis apart on sight, and reading a granulating wound or a healing fracture as a timed sequence of events rather than a static picture.

The description template

Every answer on this bench, gross or microscopic, is built the same way. Learn the order once and it works for every specimen you will ever be handed.

○ GROSS SPECIMEN, IN ORDER

- 1 ● **Organ:** name it.
- 2 ● **Size:** measured or estimated dimensions.
- 3 ● **Surface:** capsule, serosa, or skin appearance.
- 4 ● **Cut surface:** what the lesion looks like once sectioned.
- 5 ● **Colour:** named precisely, not just "abnormal."
- 6 ● **Consistency:** dry, cheesy, liquid, firm, friable.

NET RESULT a description a colleague could redraw from your words alone.

○ SLIDE, IN ORDER

- 1 ● **Architecture:** is the normal structure still recognizable, faintly outlined, or gone?
- 2 ● **Cells:** nuclei present, faded, or absent; which cell types are infiltrating.
- 3 ● **Key feature:** the one finding that names the lesion.

NET RESULT a diagnosis you can defend feature by feature, not guess by pattern-matching alone.

LAB TECHNIQUE · WRITE IT LIKE THIS

"The cut surface of the spleen shows a triangular, pale, dry lesion with a slightly elevated capsule, base toward the capsule, rimmed by a congested, hemorrhagic zone." Every clause answers one step of the template above, in order. That sentence alone should let an examiner picture the specimen before seeing it.

PART TWO

02

The Normal Baseline

Every lesion in Part Three is read against a normal you must already know cold: the spleen's own architecture, and the skin's own layers. Ten seconds spent confirming "yes, this still looks like spleen" is what tells you a lesion is focal rather than everywhere.

Normal spleen

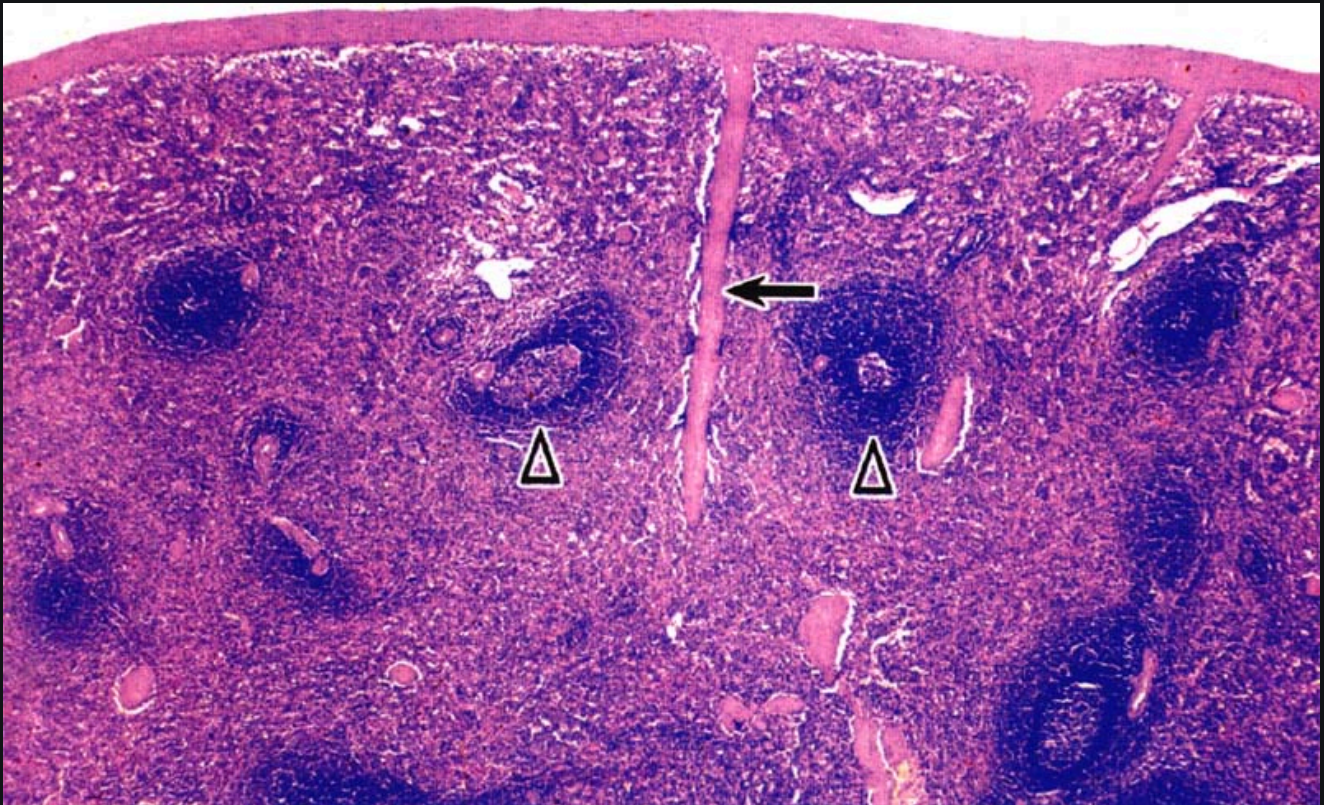
TABLE · SPLEEN AT A GLANCE

Feature	Value
Weight	110 to 200 g
Site	Left abdominal cavity, under the diaphragm
Size	10 to 12 cm × 6 to 8 cm × 3 to 4 cm
Function	The largest lymphatic organ; immune response, and filtering aged red cells

The spleen is a parenchymal organ wrapped in a dense connective-tissue capsule. Trabeculae carry trabecular arteries and veins inward from the capsule, branching into the pulp: **white pulp** (lymphoid nodules around a central artery, the immune-response tissue) and **red pulp** (a dense network of splenic sinuses, the filtering and blood-storage tissue).



Fig. 1 Normal spleen: smooth, intact capsule, uniform deep red-purple colour, firm but not hard.



MICRO **Fig. 2** Normal spleen, H&E. Round lymphoid nodules (white pulp, arrowheads) surround central arterioles; a trabecular artery runs through connective tissue (arrow) between them, with red pulp filling the rest of the field.

WHY THIS MATTERS LATER

Spleen anemic infarction (slide No.6, Part Three) is read by comparing an infarcted zone against exactly this normal picture: the same white-pulp nodules, but compressed and atrophic, sitting beside tissue that has lost its nuclei entirely.

Normal skin

Skin is built in three named layers, each with its own sublayers: the **epidermis** (keratinized stratified squamous epithelium), the **dermis** (dense connective tissue carrying vessels, nerves, and appendages), and the **subcutaneous tissue** (adipose and loose connective tissue anchoring the skin to deeper structures). The appendages, hair follicles, sweat glands, and sebaceous glands, all sit rooted in the dermis or subcutis but open onto the epidermal surface.

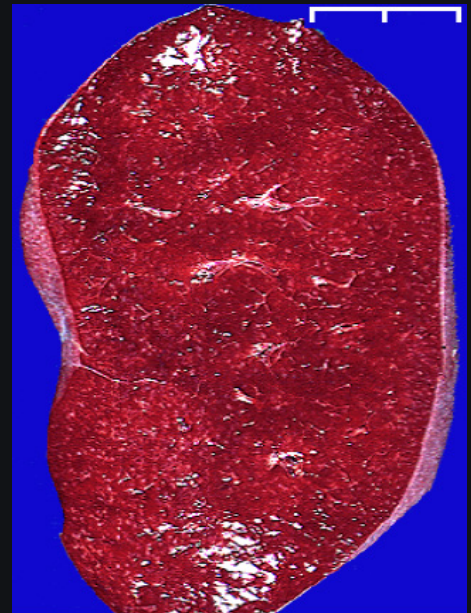
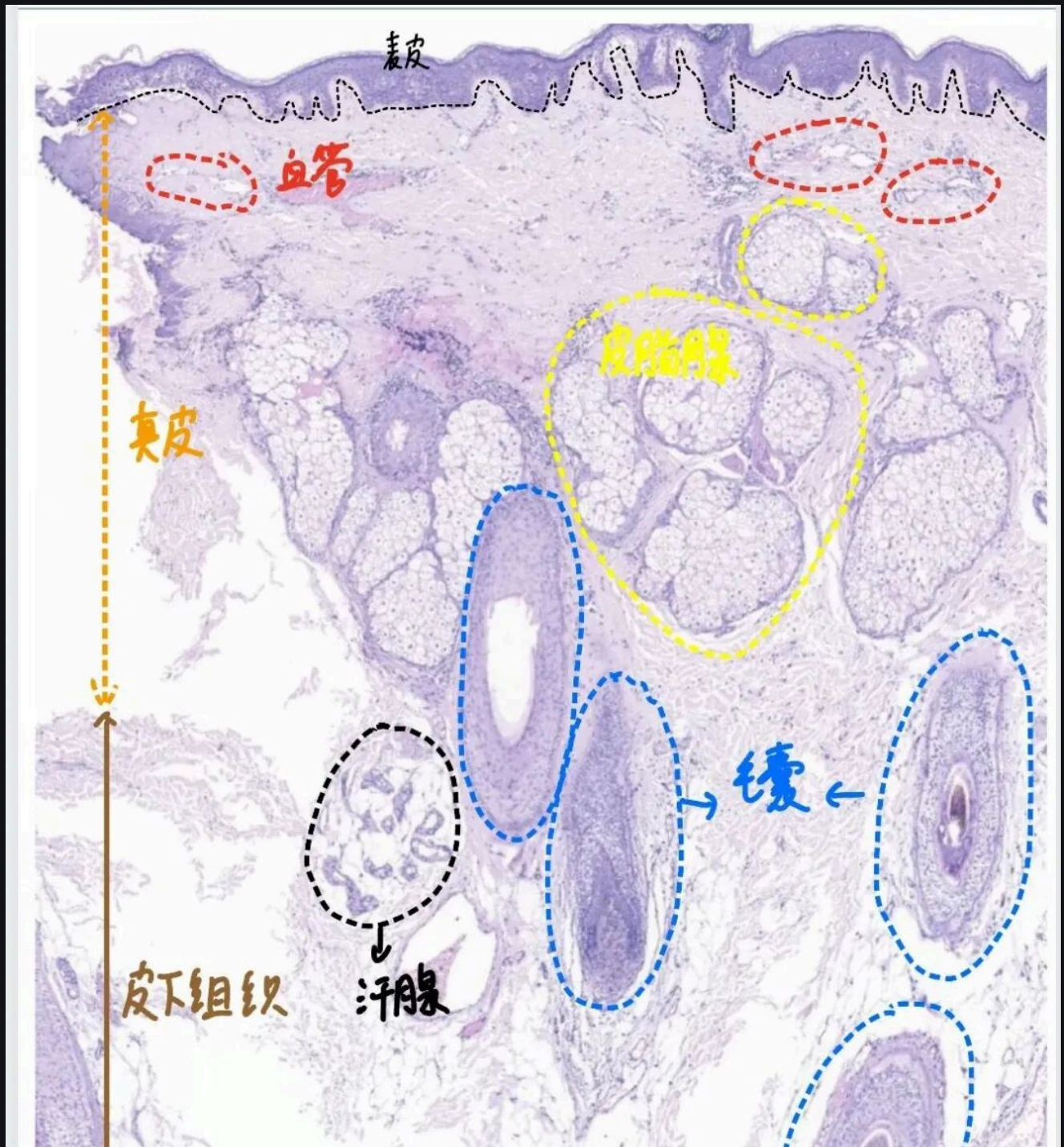


Fig. 3 Normal skin: smooth, intact surface, uniform colour, even thickness.



MICRO **Fig. 4** Normal skin, H&E, low power. Annotated on the original slide (translated here): red = blood vessel; orange arrow = dermis; brown arrow = subcutaneous tissue; black = sweat gland; yellow = sebaceous gland; blue = hair follicle.

TABLE · THE EPIDERMIS, OUTSIDE IN

Layer	Feature
Stratum corneum	Anucleate, keratinized dead cells; the barrier layer
Stratum lucidum	Thin, translucent; present only in thick (palm/sole) skin
Stratum granulosum	Keratohyalin granules mark the start of keratinization
Stratum spinosum	Desmosome-rich "prickle cells," several layers thick
Stratum basale	Single layer of proliferating cells on the basement membrane

WHY THIS MATTERS LATER

Granulation tissue (slide No.20, Part Three) forms exactly where this normal layering has been breached: the epidermis is gone at the ulcer floor, and what you see instead is new capillaries and fibroblasts filling the gap the dermis and epidermis used to occupy.

PART THREE

03

This Session's Specimens

Seven gross specimens and three key slides, in the order the bench teaches them: the three necrosis patterns first, then the two gangrenes, then repair.

Necrosis: definitions and the three morphologic patterns

DEFINITION

Necrosis is the morphologic pattern of cell death in living tissue, following irreversible injury: the cell's own hydrolytic enzymes and those of infiltrating leukocytes progressively digest it, and its contents leak out, triggering local inflammation.

Three morphologic patterns cover almost every necrosis you will see on a bench or a board exam. All three follow the same two-step mechanism, denaturation of structural proteins and enzymes versus enzymatic digestion of the dead cell, in different proportions.

TABLE · THE THREE NECROSIS PATTERNS

Pattern	Mechanism	Looks like	Classic setting
Coagulative	Protein denaturation dominates; enzymatic digestion is delayed, so the cell's outline survives	Firm, pale, dry; architecture (a "ghost" of the tissue) preserved for days	Infarction in most solid organs (heart, kidney, spleen), except the brain
Caseous	A distinct variant of coagulative necrosis: the dead cells fully lose their outline, leaving a friable, cheese-like debris	Soft, granular, yellow-white, "cheese-like"; often walled off by granulomatous inflammation	Tuberculosis, classically
Liquefactive	Enzymatic digestion dominates; hydrolytic enzymes from the dead cells or from neutrophils digest the tissue completely	Soft, liquid, viscous fluid, forming a cavity	Bacterial infection (abscess); hypoxic death in the brain

EXAM TRAP

Caseous necrosis is not a fourth, separate category: it is coagulative necrosis carried further, distinguished from ordinary coagulative necrosis by the complete loss of cellular and tissue outline (grossly and microscopically) rather than the preserved "ghost" architecture typical of a plain infarct.

· No.13* Coagulative necrosis of the spleen

No.13* Coagulative necrosis of the spleen

GROSS



ORGAN: SPLEEN

GROSS The cut surface of the spleen shows a triangular, pale, dry necrotic lesion with a slightly elevated capsule; the base is oriented toward the capsule; the periphery of the necrotic lesion is a congested, hemorrhagic zone.

Diagnosis: coagulative necrosis of the spleen (spleen infarct)

○ WHY A WEDGE, WHY PALE

- 1 The spleen's segmental arteries are functional end-arteries; occlusion of one (classically an embolus) cuts blood supply to the whole wedge of tissue it alone feeds.

- 2 The wedge points toward the hilum (the vessel's origin) and its base sits at the capsule (the vessel's furthest reach), which is why the base, not the apex, sits against the capsule.
- 3 Because the spleen is a solid organ with little collateral flow at the segmental level, the result is **pale (anemic) infarction**: coagulative necrosis with little residual blood, hence the dry, pale cut surface.

NET RESULT a pale, dry, wedge-shaped lesion with its base at the capsule, exactly what the specimen shows.

· No.14 Tuberculosis ball of the lung (caseous necrosis)

No.14 Tuberculosis ball of the lung

GROSS



ORGAN: LUNG

GROSS The lung tissue shows two spherical lesions, grayish-yellow; the cheese-like necrotic material and carbon deposits form a layered, concentric structure. Some of the caseous material has fallen away into a cavity, surrounded by fibrous tissue.

Diagnosis: caseous necrosis of the lung (tuberculoma)

HIGH YIELD · WHY "CHEESE"

Mycobacterium tuberculosis triggers a cell-mediated (delayed-type hypersensitivity) immune response: activated macrophages transform into epithelioid cells and fuse into Langhans' giant cells, walling the organism off in a granuloma. At the centre of that granuloma, the combination of mycobacterial lipid content and the immune reaction produces necrotic debris that is friable and grossly resembles soft, crumbly cheese, hence "caseous," from the Latin *caseus*.

· No.15 Bacterial liver abscess (liquefactive necrosis)

No.15 Bacterial liver abscess

GROSS



ORGAN: LIVER

GROSS Several round cavities of different sizes are seen on the cut surface of the liver, larger ones about 1 × 0.5 cm and smaller ones about 0.3 cm across; the abscess wall is thick, the cavity contains yellowish-white necrotic material (pus), and the surrounding liver tissue is dark red.

Diagnosis: liquefactive necrosis of the liver (bacterial liver abscess)

○ WHY LIQUID, NOT SOLID

- 1 Pyogenic bacteria recruit massive neutrophil infiltration.
- 2 Neutrophils release potent hydrolytic enzymes (proteases, lipases) into the focus.

- 3 Those enzymes digest the dead tissue completely, faster than any protein-denaturation process can preserve it, converting solid parenchyma into a viscous, protein-rich fluid: pus.

WHY: *this is the same mechanism that liquefies hypoxic brain tissue, which is why the CNS liquefies rather than infarcting the way the spleen or heart does.*

NET RESULT a fluid-filled cavity with a fibrous wall, not a firm pale wedge.

Gangrene: wet vs. dry

DEFINITION

Gangrene is not a distinct type of necrosis; it is a clinical term for coagulative necrosis of an extremity or a hollow viscus that has become extensive enough to be grossly obvious, modified by whether bacterial infection has been superimposed.

TABLE · WET VS. DRY GANGRENE

Feature	Dry gangrene	Wet gangrene
Underlying process	Coagulative necrosis from arterial occlusion (ischemia) alone	Coagulative necrosis plus superimposed bacterial infection
Gross appearance	Dry, shrunken, dark brown to black, mummified; clear demarcation from viable tissue	Swollen, wet, foul-smelling; boundary with viable tissue unclear; rapidly progressive
Mechanism of colour	Loss of blood supply and gradual desiccation	Hemoglobin breakdown products plus bacterial putrefaction
Typical site	Lower extremity in peripheral vascular disease	Bowel, appendix; anywhere ischemic tissue meets the gut's bacterial flora

· No.16 Gangrenous appendicitis (wet gangrene)

No.16 Gangrenous appendicitis

GROSS



ORGAN: APPENDIX

GROSS The appendix is swollen and thickened; the serosal surface is scattered with grayish-yellow inflammatory exudate and shows vascular dilation and congestion. On section, the wall is thickened, and a fecalith is present in the lumen. In severely affected areas the wall is dark black with focal perforation.

Diagnosis: wet gangrene of the appendix (gangrenous appendicitis)

EXAM TRAP · PERFORATION

A perforated, dark, foul appendix is the textbook picture of wet gangrene specifically because the appendiceal lumen is already colonized by gut flora: once the wall becomes ischemic (usually from luminal obstruction by a fecalith, raising intraluminal pressure above capillary perfusion pressure), those bacteria invade dead tissue immediately, there is no "dry" phase to observe.

No.17 Wet gangrene of the small intestine**No.17 Wet gangrene of the small intestine**

GROSS



ORGAN: SMALL INTESTINE

GROSS The affected intestinal segment is dark black; the wall is thickened by congestion, edema, and hemorrhage; the boundary with normal bowel is unclear, and the lesion area has lost its normal luster. A roundworm (*Ascaris lumbricoides*) is visible at one end of the segment.

Diagnosis: wet gangrene of the small intestine (hemorrhagic infarction)

LAB TECHNIQUE · WHY "HEMORRHAGIC"

Bowel has a dual, anastomosing blood supply and loose, distensible tissue, so venous outflow obstruction (here from a mass of roundworms causing intestinal obstruction and secondary strangulation) lets blood continue pooling into the segment even as outflow fails, producing the red-black, blood-engorged appearance of a hemorrhagic (red) infarct, unlike the pale, bloodless infarct of the spleen.

· No.18* Dry gangrene of the lower extremity**No.18* Dry gangrene of the lower extremity**

GROSS



ORGAN: LOWER EXTREMITY (FOOT)

GROSS Necrosis occurs distal to the ankle joint. The necrotic area is dirty grayish-black, dry, hard, and clearly demarcated from the normal, viable tissue proximally.

Diagnosis: dry gangrene of the foot

WHY DRY, WHY SHARPLY DEMARCATED

- 1 Chronic arterial insufficiency (atherosclerosis, often in a diabetic patient) narrows flow gradually, without an accompanying venous or infective process.
- 2 Tissue distal to the occlusion undergoes coagulative necrosis and then slowly desiccates (dries out) rather than putrefying, because no bacterial invasion is driving liquefaction.
- 3 The line between dead and living tissue stays sharp because ischemia, unlike infection, does not spread actively into adjacent viable tissue.

NET RESULT a dry, hard, black extremity with a clean edge, the opposite of the blurred, wet, infected margin of gangrenous appendicitis.



Fig. 8 Close-up: the circled zone shows the sharp line between mummified black tissue and pink, viable skin proximal to it, the "clear boundary" that distinguishes dry from wet gangrene.

EXAM TRAP

Dry gangrene can convert to wet gangrene if bacteria secondarily invade the desiccated tissue (common if the patient is diabetic, since hyperglycemia impairs neutrophil function); the distinction is about infection, not about which artery was occluded.

Repair: granulation tissue and fracture healing

DEFINITION

Granulation tissue is the specialized tissue that fills a defect during healing: new capillary sprouts (angiogenesis) plus proliferating fibroblasts, in a loose, edematous, myxoid stroma infiltrated by inflammatory cells. Its pink, granular, translucent gross appearance gives it the name.

GRANULATION TISSUE, IN ORDER

- 1 New capillaries sprout from the wound margin (angiogenesis), driven chiefly by VEGF.
 - 2 Fibroblasts migrate in and proliferate, laying down loose type III collagen and ground substance.
 - 3 Macrophages continue to clear debris and secrete growth factors that sustain both processes.
 - 4 Over weeks, fibroblasts differentiate into myofibroblasts that contract the wound, and type III collagen is remodelled into stronger type I collagen, forming mature scar.
- WHY:** granulation tissue is a temporary scaffold; its whole job is to convert itself into scar and then largely disappear.

NET RESULT a temporary, vascular, fibroblast-rich tissue that matures into permanent, avascular scar.

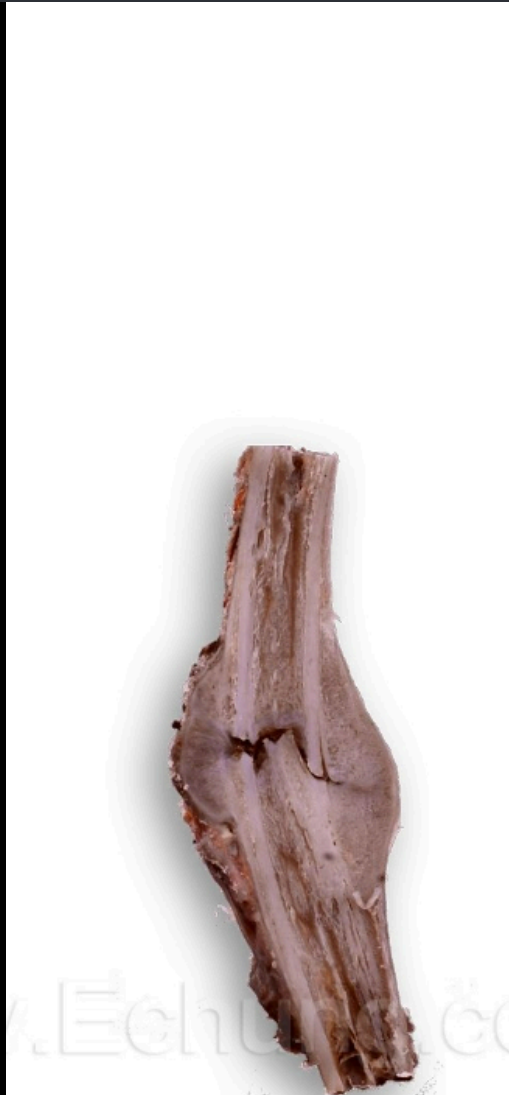
Fracture repair follows the identical logic, on a scaffold of bone rather than skin: a hematoma forms first, organizes into granulation tissue, then a soft (fibrous/cartilaginous) callus bridges the fracture, and finally that soft

callus is remodelled into a hard, bony callus.

· No.23 Fibrous callus formation (tubular bone)

No.23 Fibrous callus formation

GROSS



ORGAN: LONG TUBULAR BONE

GROSS A fracture of the long tubular bone at its mid-shaft, with a spindle-shaped expansion (fibrous callus) around the fracture site. The callus is proliferated by a large amount of fibrous tissue, becoming progressively organized and then fibrosed.

Diagnosis: fibrous callus formation

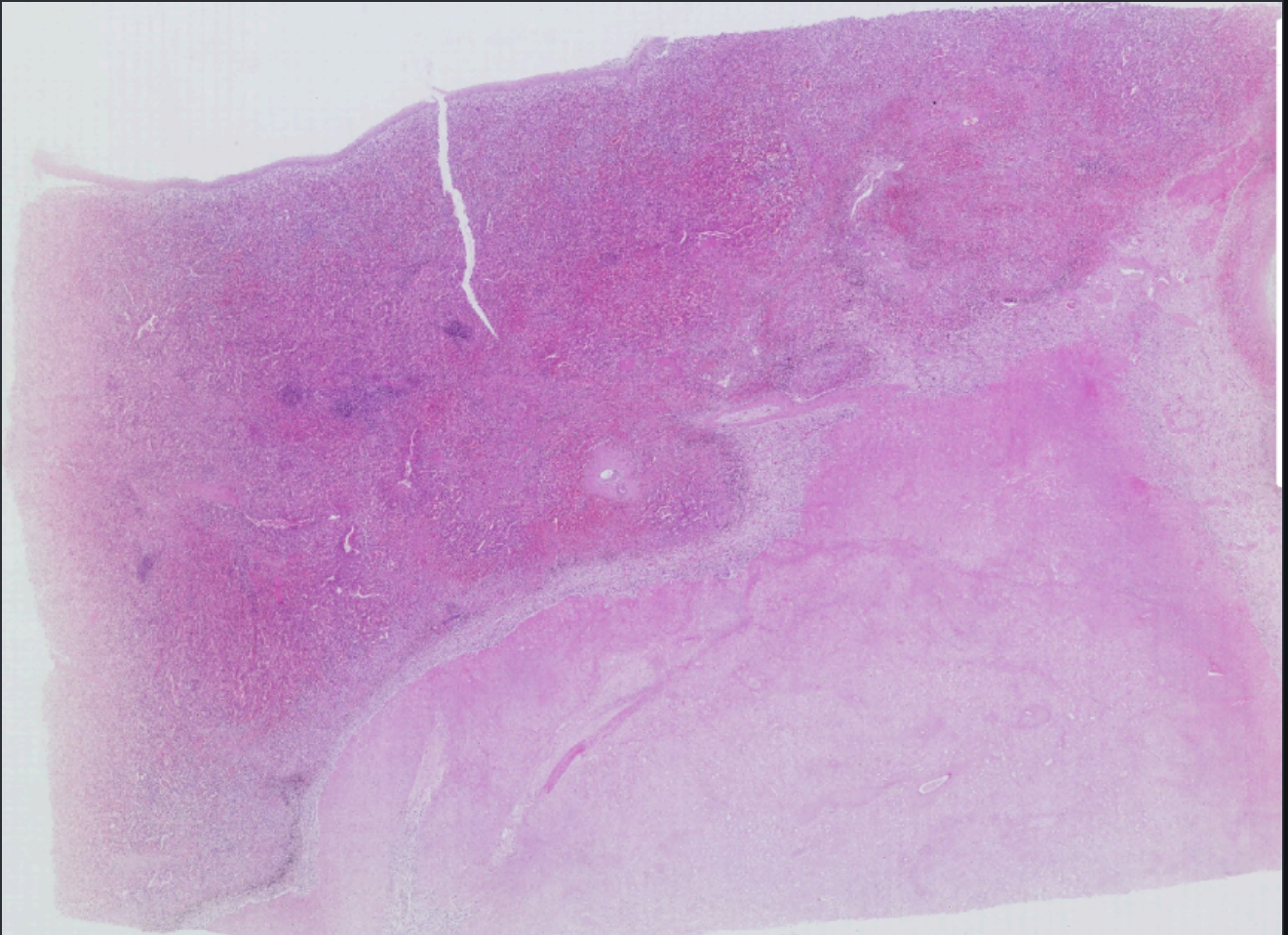
TABLE · FRACTURE REPAIR, STAGE BY STAGE

Stage	What forms
Hematoma	Blood clot fills the fracture gap within hours
Soft callus	Granulation tissue, then fibrous/cartilaginous callus, bridges the fragments (this specimen)
Hard (bony) callus	The soft callus is mineralized into woven bone
Remodelling	Woven bone is remodelled into lamellar bone along stress lines, callus shrinks toward normal contour

Slide No.6* Spleen anemic infarction

No.6* Spleen anemic infarction

MICRO



ORGAN: SPLEEN

MICRO In the irregular infarcted area of the spleen, nuclei have disappeared and only a faint outline of the splenic structure remains. Inflammatory cells infiltrate around the infarction, forming an inflammatory response zone, accompanied by granulation tissue. Splenic sinus congestion is obvious, and the white pulp (lymphoid nodules) is compressed and atrophic.

Diagnosis: spleen anemic infarction (coagulative necrosis of the spleen)

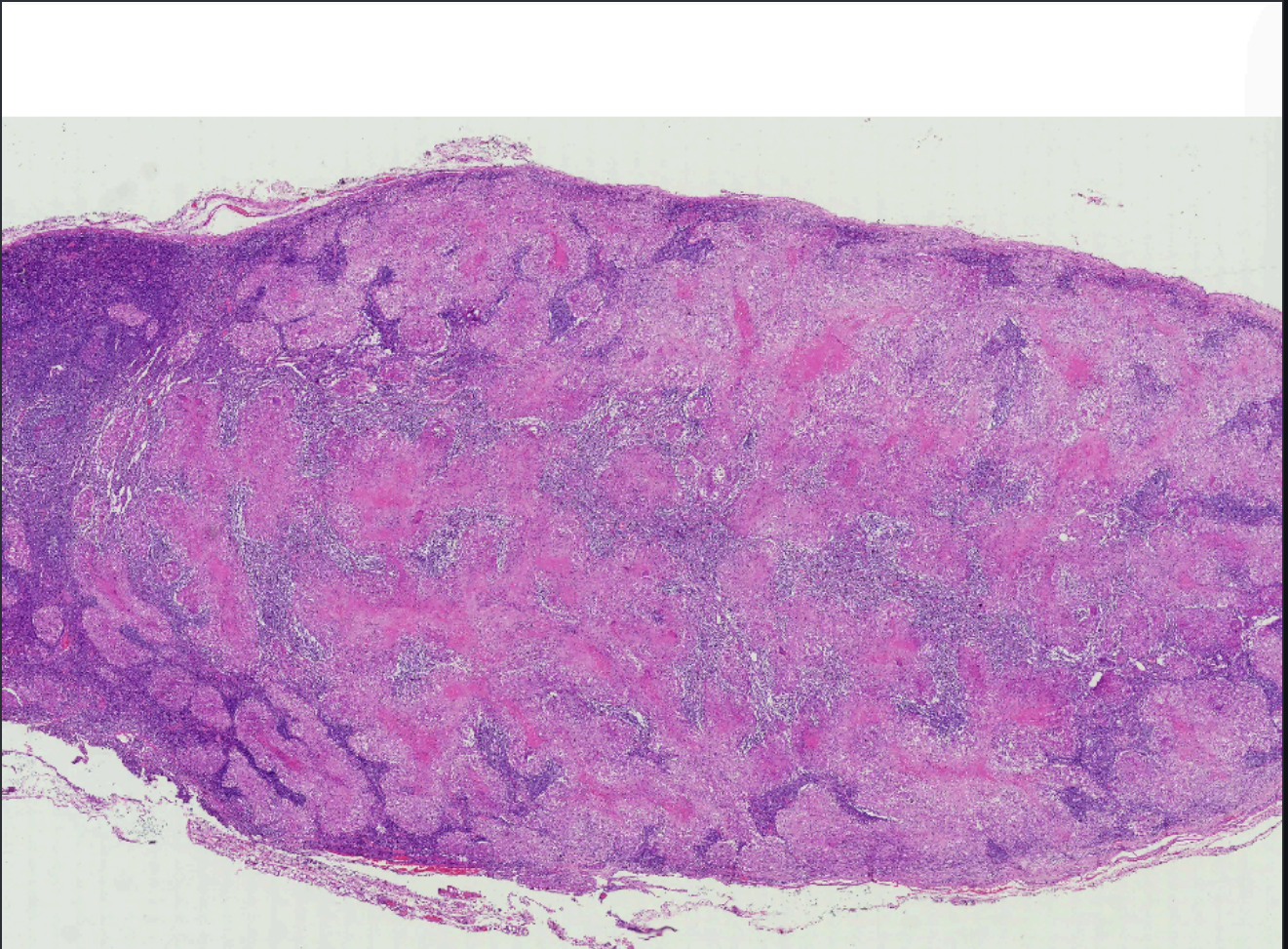
READING IT AGAINST THE BASELINE

Compare directly with Fig. 2 (Part Two): the same round white-pulp nodules are here, but shrunk and compressed rather than plump, and the surrounding red pulp has lost its nuclei entirely, exactly the "ghost architecture" that defines coagulative necrosis.

Slide No.86 Caseous necrosis in lymph node

No.86 Caseous necrosis (lymph node tuberculosis)

MICRO



ORGAN: LYMPH NODE

MICRO Numerous nodular lesions (tuberculous nodules, granulomas) are seen throughout the lymph node. At high magnification each nodule is composed of short spindle cells (epithelioid cells) in irregular arrangement, with scattered Langhans' giant cells. Many nodules show focal caseous necrosis (fine granular, red-staining material) at their centre.

Diagnosis: caseous necrosis, lymph node tuberculosis

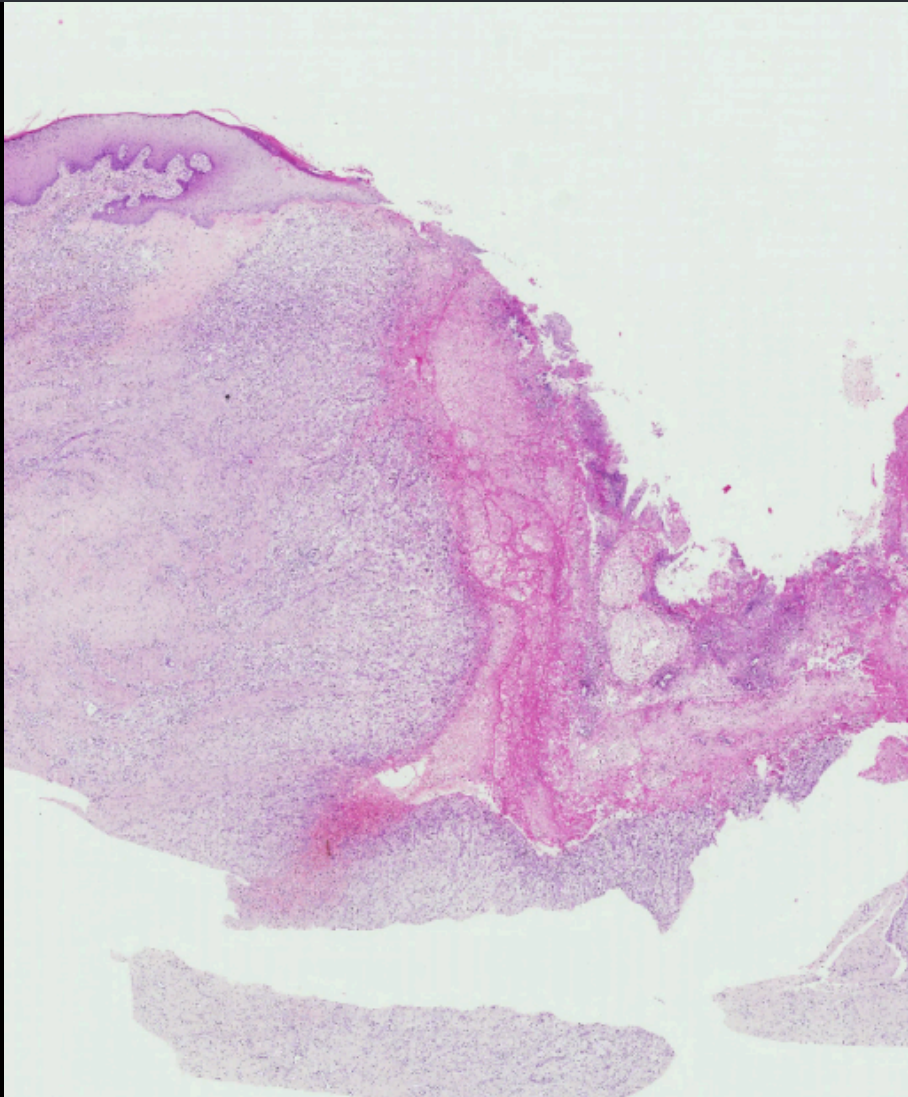
HIGH YIELD · THE GRANULOMA'S OWN ANATOMY

A tuberculous granuloma has a signature layout worth naming layer by layer: a central zone of caseous necrosis, surrounded by epithelioid macrophages and Langhans' giant cells, surrounded in turn by a rim of lymphocytes and fibroblasts. This same specimen pairs directly with Gross No.14 (tuberculoma): the microscope explains why the gross lesion is cheese-like and layered.

- **Slide No.20* Healing by second intention (granulation tissue)**

No.20* Granulation tissue (skin ulcer)

MICRO



ORGAN: SKIN

MICRO At the base of the ulcer, necrotic tissue (fine red granular material) and inflammatory exudate, including inflammatory cells and fibrin, form a layer of necro-inflammatory debris; beneath it lies abundant granulation tissue. The granulation tissue is loose, with edematous stroma and pale staining; numerous new capillaries and fibroblasts are seen, with scattered inflammatory cells infiltrating throughout.

Diagnosis: granulation tissue, skin ulcer healing by second intention

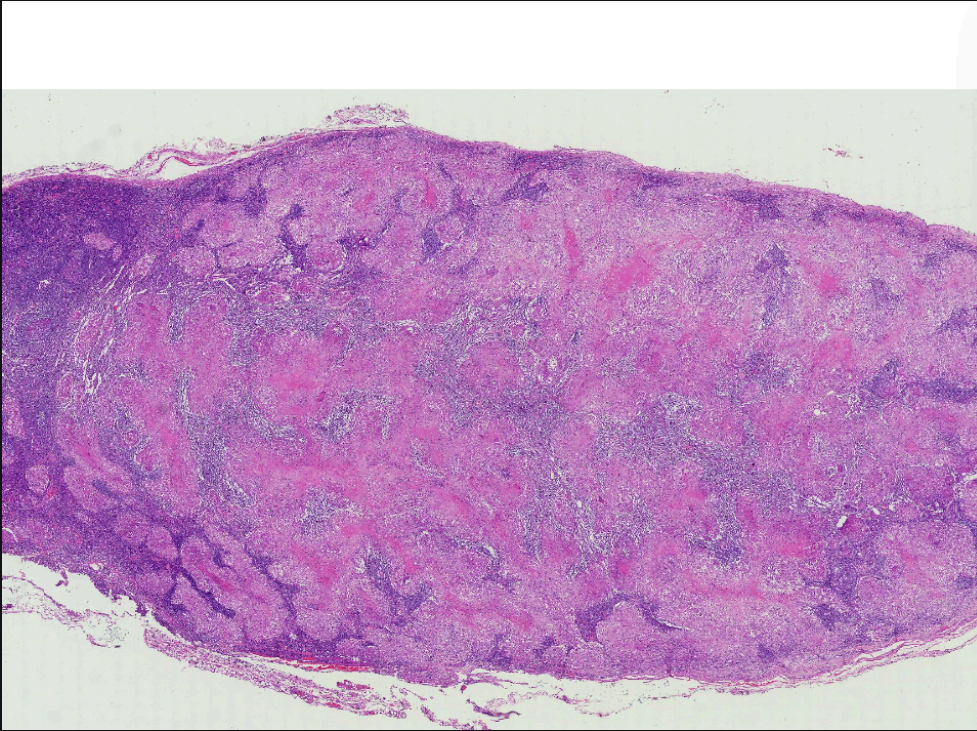
LAB TECHNIQUE · FIRST VS. SECOND INTENTION

Healing by **first intention** closes a clean, apposed incision with minimal granulation tissue and a fine scar. Healing by **second intention**, seen here, closes a large, open defect (an ulcer, a wide wound) that cannot be directly apposed: it needs substantially more granulation tissue to fill the gap, more wound contraction by myofibroblasts, and leaves a larger scar.

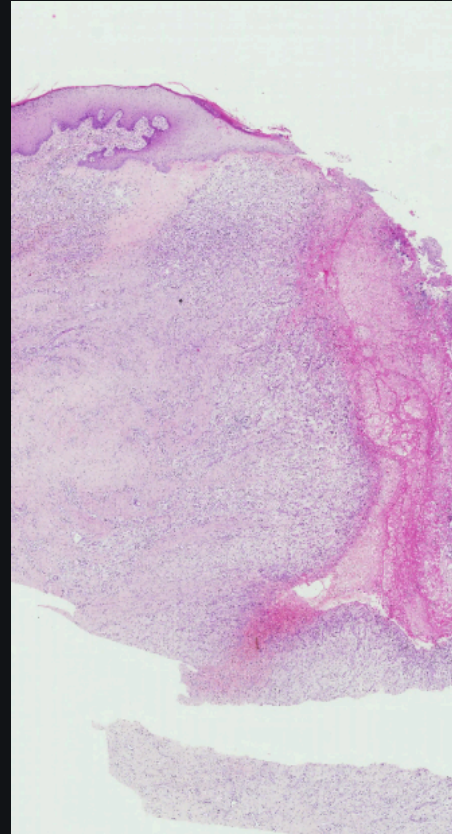
Spot-exam practice

Cover the diagnosis lines above and name each specimen from the image and number alone; check yourself against the answer key at the end.

**No.13*****No.18***



Slide No.86



Slide No.20*

REVIEW

R

Homework, Self-Test, Glossary

The homework set as given, a ten-question spot-exam-style bank with an answer key, an identification checklist for last-minute review, and a bilingual glossary.

Homework

AS GIVEN

1. Describe the gross specimens: (1) No.13* coagulative necrosis of the spleen; (2) No.18* dry gangrene of the lower extremity.
2. Describe the slides: (1) No.6* caseous necrosis in lymph node; (2) No.20* healing by second intention of skin (granulation tissue).

NOTE ON THE HOMEWORK'S OWN NUMBERING

The homework sheet asks for slide "No.6* caseous necrosis in lymph node," but the specimen list and this session's own slide deck give No.6* as the **spleen anemic infarction** slide, and No.86 as the lymph-node caseous necrosis slide. Answer using the correct specimen for each diagnosis (No.6* spleen infarction, No.86 lymph node), not by number alone, and confirm with your instructor which numbering the graded submission expects.

Self-test question bank

Q1

A spleen specimen shows a firm, pale, dry, wedge-shaped lesion with its base at the capsule. The most likely diagnosis is:

Q2

Caseous necrosis is best described as:

Q3

A liver specimen shows multiple round cavities filled with yellow-white fluid, thick walls, and dark red surrounding parenchyma. Which necrosis pattern is this?

Q4

Name the mechanistic difference between coagulative and liquefactive necrosis.

Q5

A gangrenous appendix is black, foul-smelling, and perforated, with an unclear boundary to viable tissue. Is this wet or dry gangrene, and why?

Q6

What clinical process converts dry gangrene into wet gangrene?

Q7

On a lymph node slide, epithelioid cells and scattered Langhans' giant cells surround a focus of granular, red-staining necrotic debris. Name the lesion and the organism it most classically implies.

Q8

List the two components that define granulation tissue at the cellular level.

Q9

Order the four stages of fracture repair.

Q10

Why does healing by second intention leave a larger scar than healing by first intention?

Answer key

A1 Coagulative necrosis (a pale/anemic splenic infarct following segmental artery occlusion).

A2 A variant of coagulative necrosis, classically from tuberculosis, in which the dead tissue loses its cellular and tissue outline entirely, leaving friable, cheese-like debris.

A3 Liquefactive necrosis (bacterial liver abscess): neutrophil enzymes digest the tissue completely into pus.

A4 Coagulative necrosis: protein denaturation dominates and the cell's ghost outline survives. Liquefactive necrosis: enzymatic digestion dominates and the tissue is dissolved completely.

A5 Wet gangrene: bacterial invasion of the already-necrotic wall (from the appendix's own luminal flora) drives putrefaction, foul odour, and an unclear, actively spreading margin.

A6 Secondary bacterial infection of the desiccated, dead tissue (more likely in a diabetic patient, whose impaired neutrophil function permits bacterial invasion).

A7 A tuberculous granuloma with central caseous necrosis; classically implies infection with *Mycobacterium tuberculosis*.

A8 New capillaries (angiogenesis) and proliferating fibroblasts, in a loose, edematous stroma.

A9 Hematoma, then soft (fibrous/cartilaginous) callus, then hard (bony) callus, then remodelling.

A10 A large, open defect cannot be directly apposed, so it needs substantially more granulation tissue to fill the gap and more myofibroblast-driven contraction, leaving more collagen behind as scar.

Identification checklist

- ☐ Coagulative necrosis: pale, firm, dry, architecture (ghost outline) preserved
- ☐ Caseous necrosis: soft, granular, yellow-white, cheese-like, outline lost entirely
- ☐ Liquefactive necrosis: soft, liquid, cavity-forming
- ☐ Dry gangrene: dry, mummified, sharply demarcated, no infection
- ☐ Wet gangrene: wet, foul, unclear margin, infected
- ☐ Spleen anemic infarction: white-pulp nodules compressed, red pulp anucleate
- ☐ Lymph node caseous necrosis: epithelioid cells, Langhans' giant cells, central caseation
- ☐ Granulation tissue: new capillaries, fibroblasts, loose edematous stroma, inflammatory cells
- ☐ Fibrous callus: spindle-shaped swelling at a fracture site, proliferating fibrous tissue

Glossary

Caseous necrosis

A variant of coagulative necrosis with complete loss of cellular and tissue outline, leaving friable, cheese-like debris; classically tuberculosis.

Coagulative necrosis

Protein denaturation dominates; the cell's outline survives as a "ghost" for days; the pattern of most solid-organ infarcts.

Dry gangrene

Coagulative necrosis of an extremity from ischemia alone, without infection; dry, mummified, sharply demarcated.

Fecalith

A hardened mass of fecal material that can obstruct the appendiceal lumen.

Gangrene

A clinical term for extensive coagulative necrosis of an extremity or hollow viscus, subclassified as wet or dry by the presence of superimposed infection.

Granulation tissue

New capillaries plus proliferating fibroblasts in a loose, edematous stroma; the temporary scaffold of healing.

Hemorrhagic infarction

A red, blood-engorged infarct typical of loose tissue with dual blood supply, such as bowel; contrasted with the pale infarct of a solid organ.

Langhans' giant cell

A multinucleated cell formed by fusion of epithelioid macrophages, nuclei arranged in a peripheral horseshoe; a hallmark of the tuberculous granuloma.

Liquefactive necrosis

Enzymatic digestion dominates; the tissue is dissolved into a viscous fluid; typical of bacterial abscess and hypoxic brain death.

Necrosis

The morphologic pattern of cell death in living tissue following irreversible injury.

Second intention (healing by)

Closure of a large, open wound that cannot be directly apposed, requiring substantial granulation tissue and wound contraction.

Wet gangrene

Coagulative necrosis with superimposed bacterial infection; wet, foul-smelling, with an unclear, actively spreading margin.